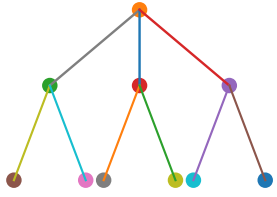
The complete generative deformation object

expand quotient complete certify

Seed and local fiber

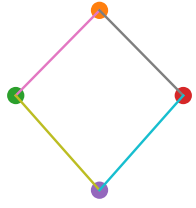
x_0 with its own admissible directions

Raw route forest



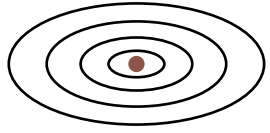
all finite direction words with route multiplicity

Quotient atlas



relations merge equivalent endpoints, not histories

Completed boundary



proved limits of compatible infinite branches

Certified families

parameters, closure laws, relations, invariants

Route data, object identity, convergence data, and family structure remain distinct and auditable.